

BAAZ JKBOSE



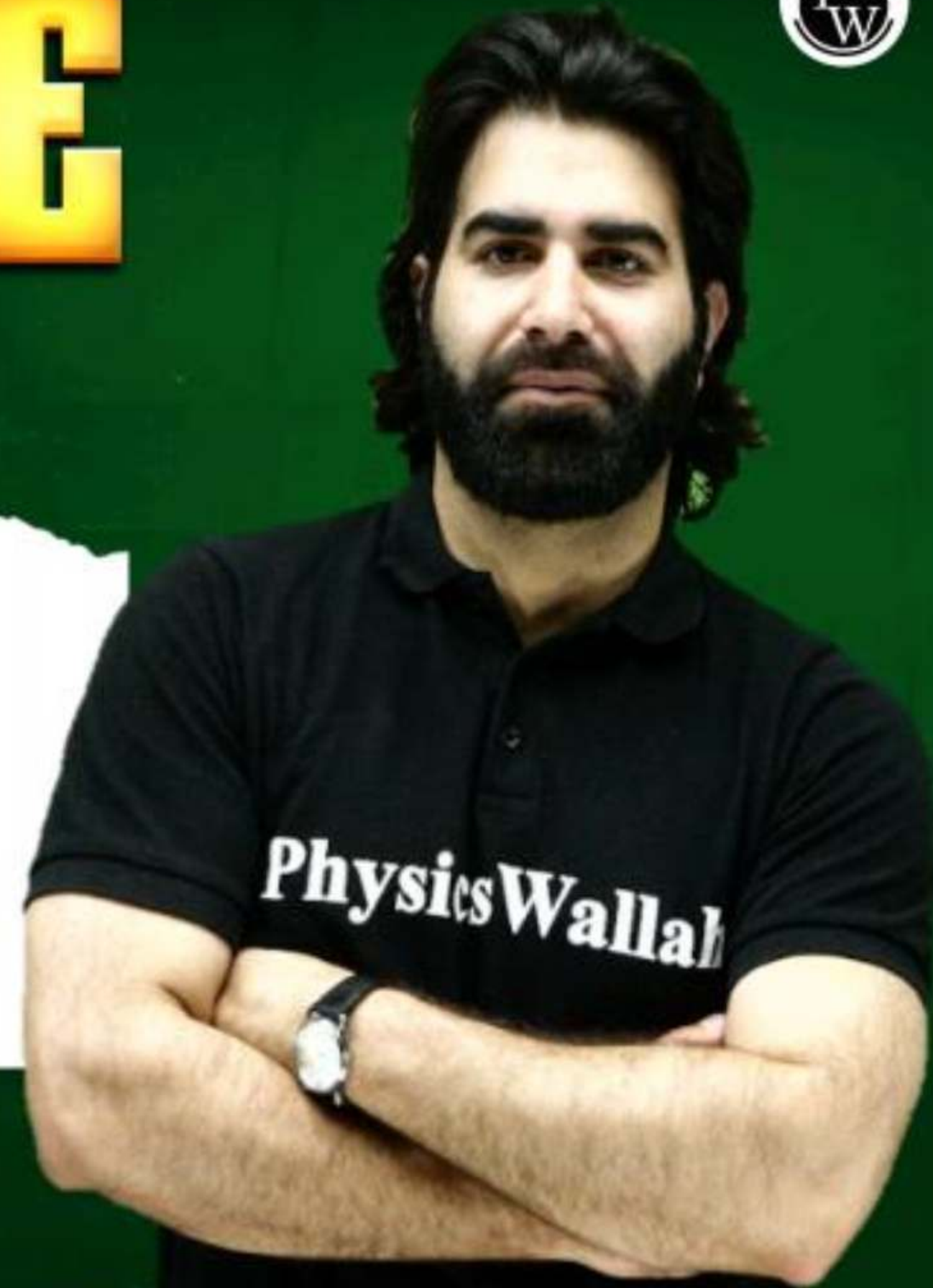
2026 CLASS-11TH

Locomotion and
Movement

LECTURE- 6

Zoology

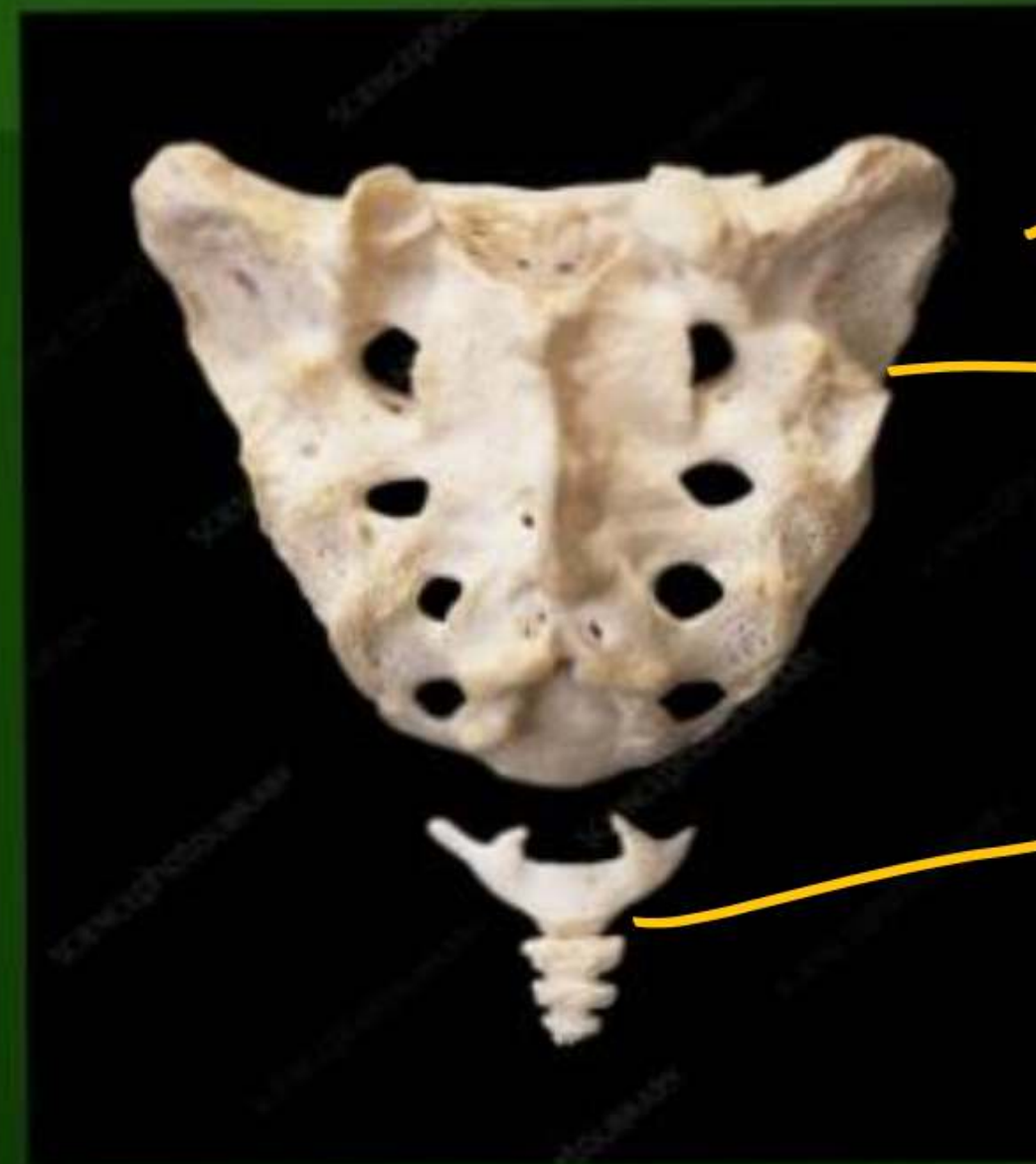
Dr Adil Altaf sir



Topics To Be Covered

1 Skeletal System

2. Human Skeletal System



Guess the Bone

Sacrum

Coccyx

→ Ribs & Sternum

→ Ribs

- Bicephalic

- Fused on Both sides
- Dorsally - vertebral column.
- Ventrally - Sternum.

→ 12 pairs (24).

→ First seven pairs (14)

- True Ribs

- Vertebrosteral Ribs

→ 8th, 9th, 10th Pair

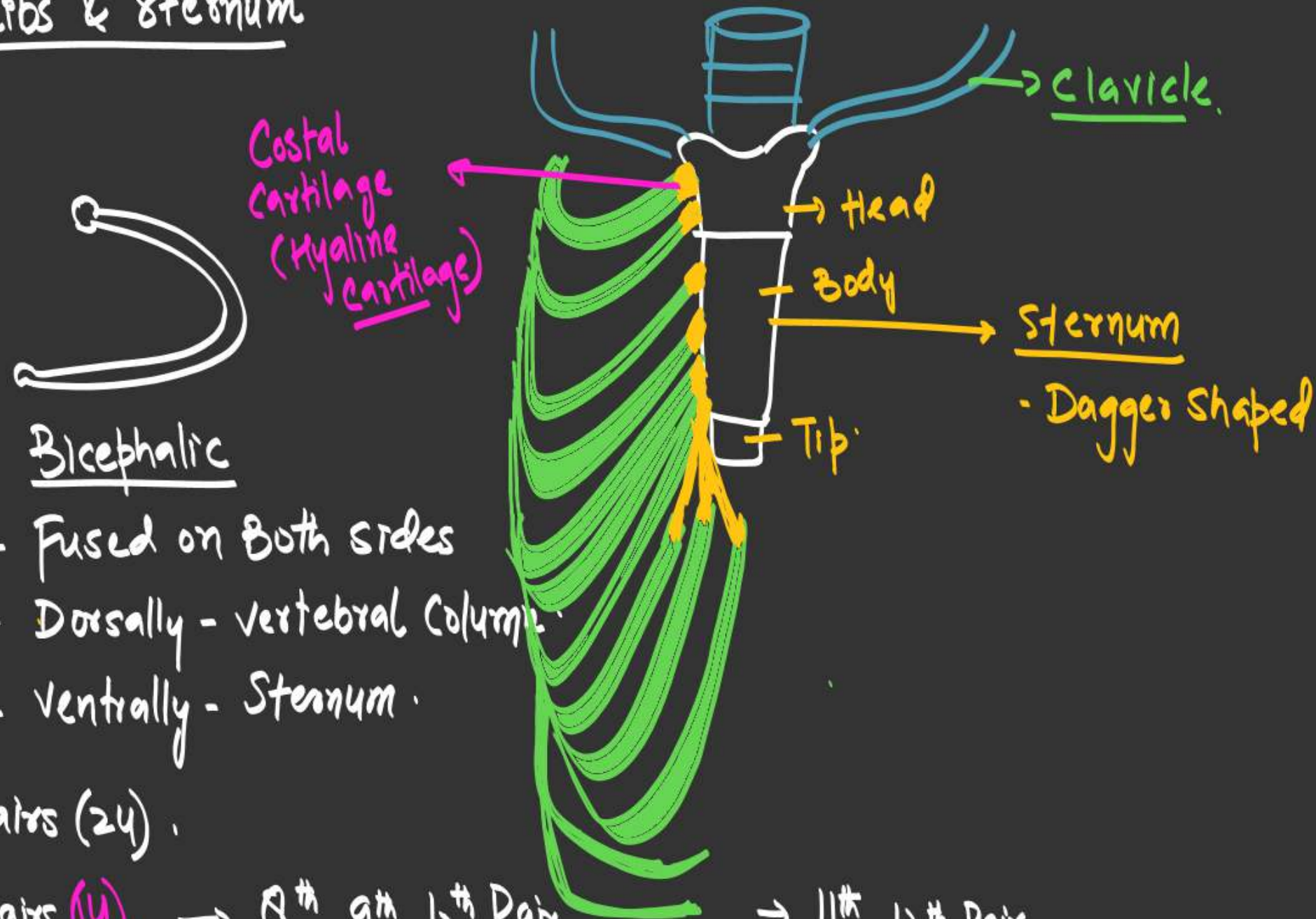
→ False Ribs.

→ Vertebrochondral Ribs

→ 11th, 12th Pair

- Floating Ribs

- Not Attached with Sternum.



→ Appendicular Skeleton (126 Bones)

① Pectoral Girdle

- To it upper Limbs / Arms Attached.
- It has two halves,
- Each Half has 2 Bones.
- Total 4 Bones.

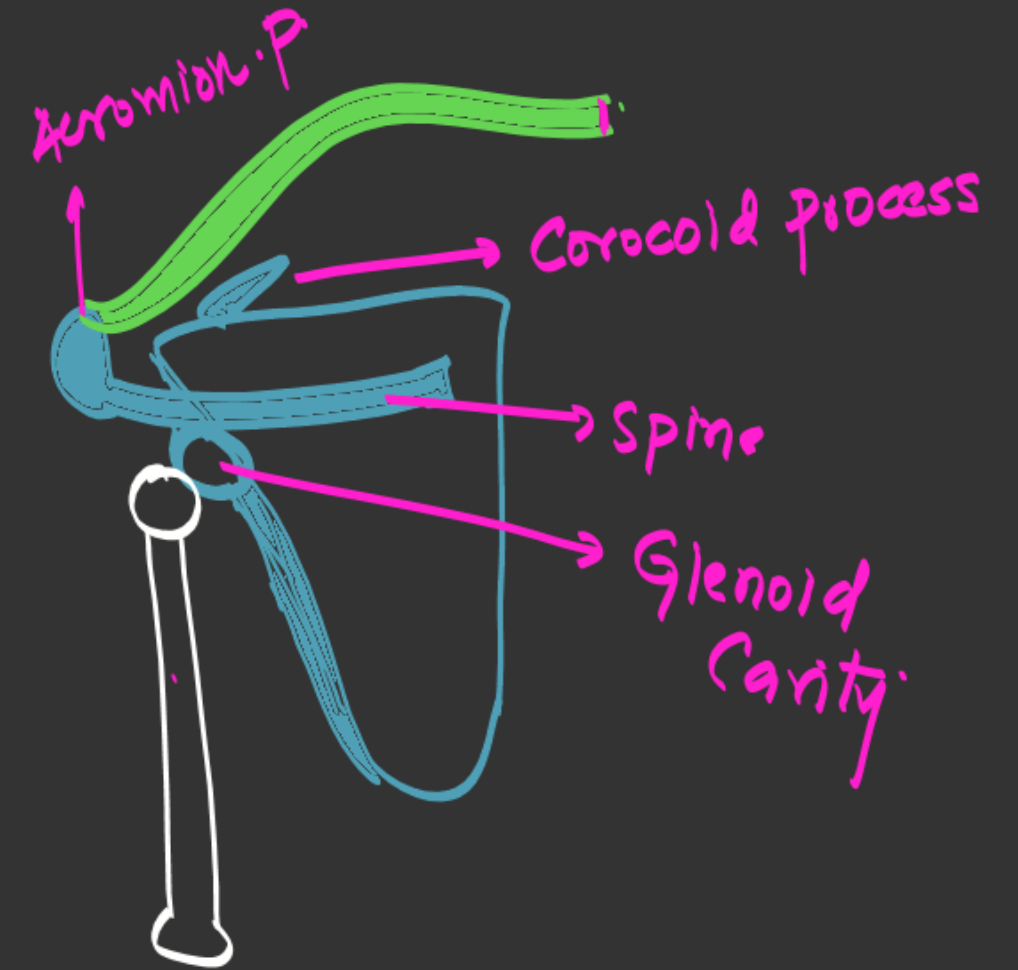


② Clavicle

- Curved but long Bone
- Also k/as Collar Bone.
- Forms joint with Scapula & Sternum.

→ Scapula.

- It is a triangular Flat Bone
- It has an elevated Ridge k/as Spine of Scapula.
- Spine Bears Acromion process; to which clavicle
- Attachs.
- outer Margin Bears a cavity = Glenoid cavity, to it head of Humerus Articulates.



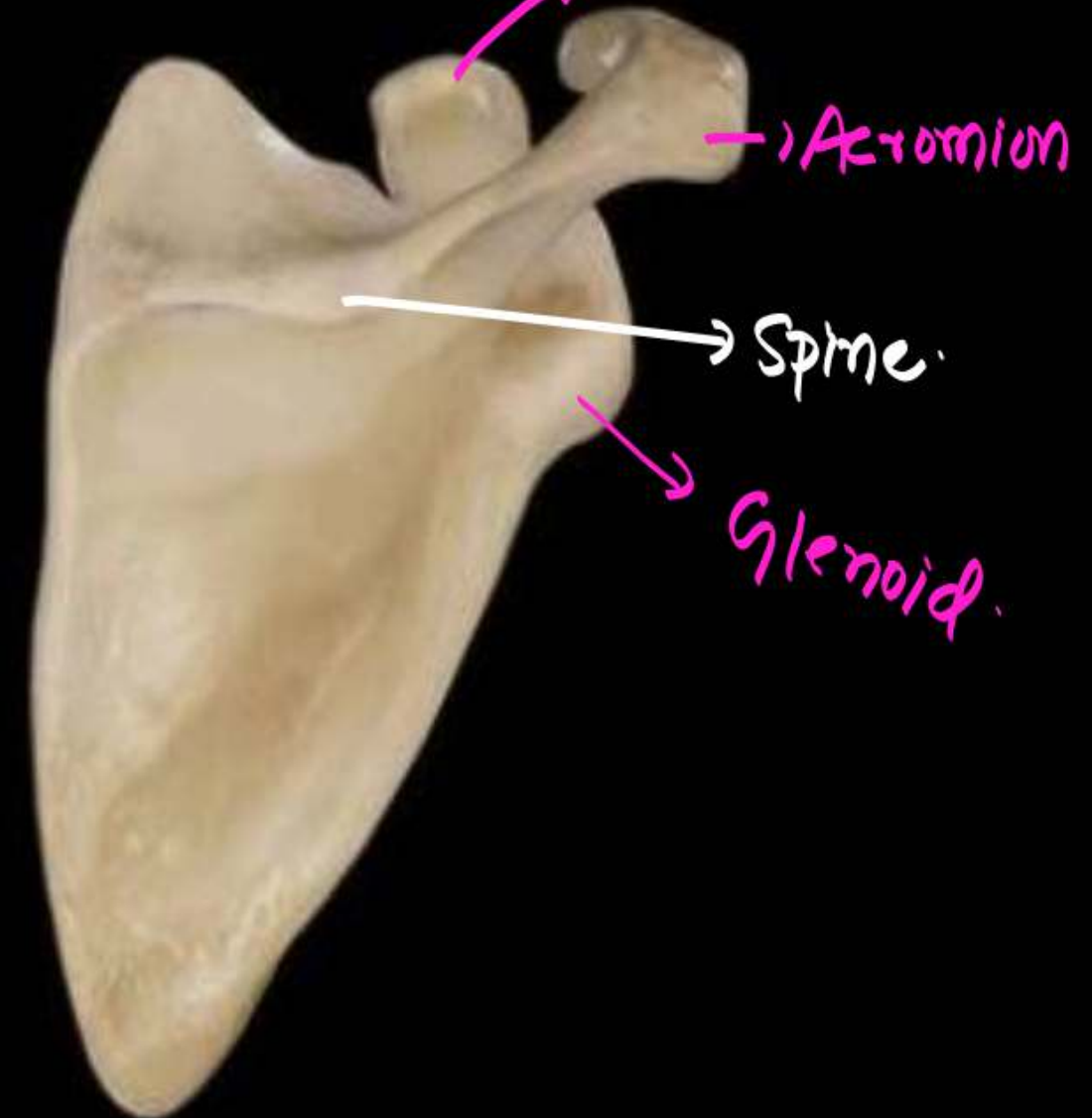
Upper Limb.

→ 9+ has

Inner Side

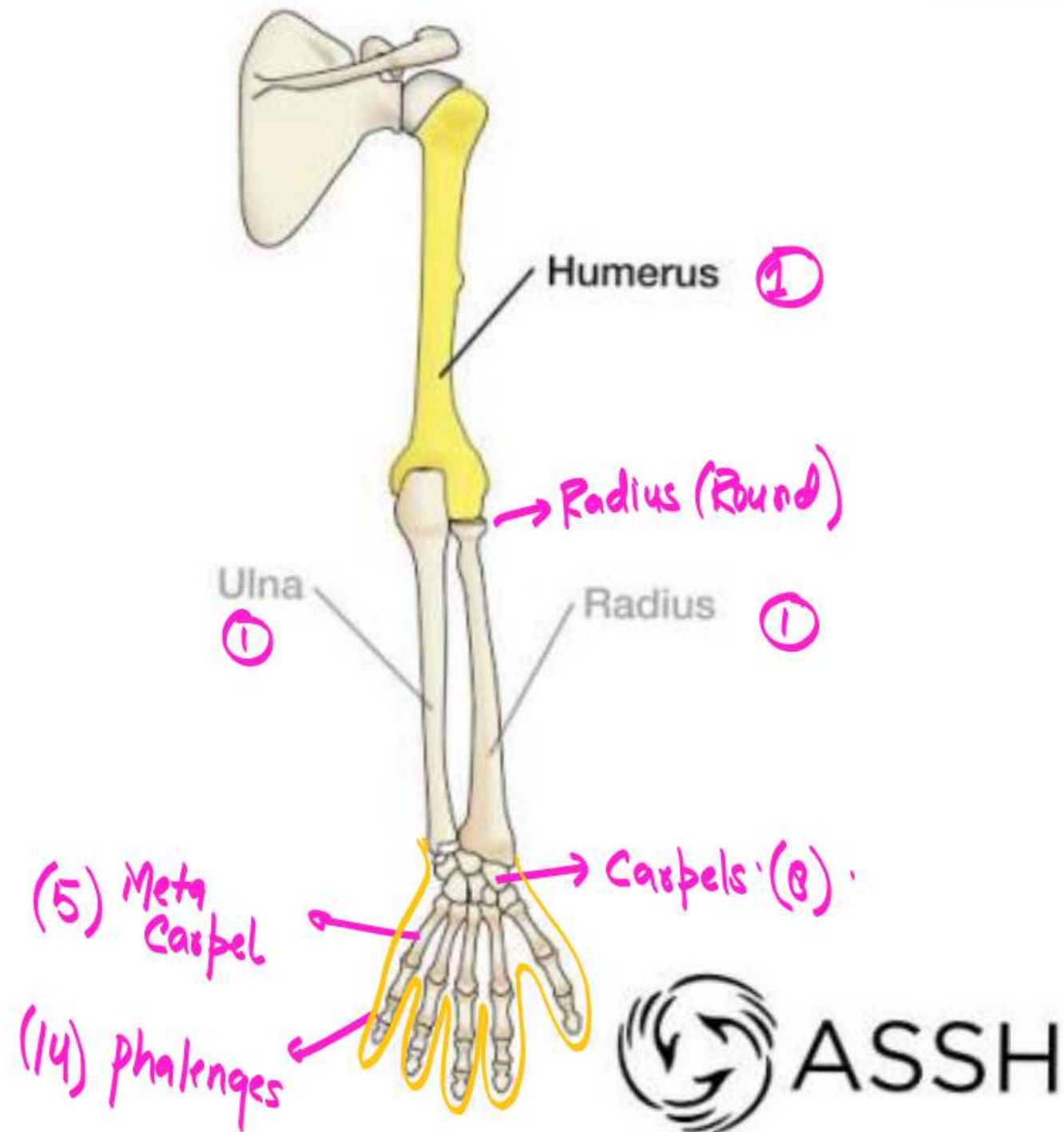


outer side



→ Upper Limb

→ It has 30 Bones



- Pelvic Girdle

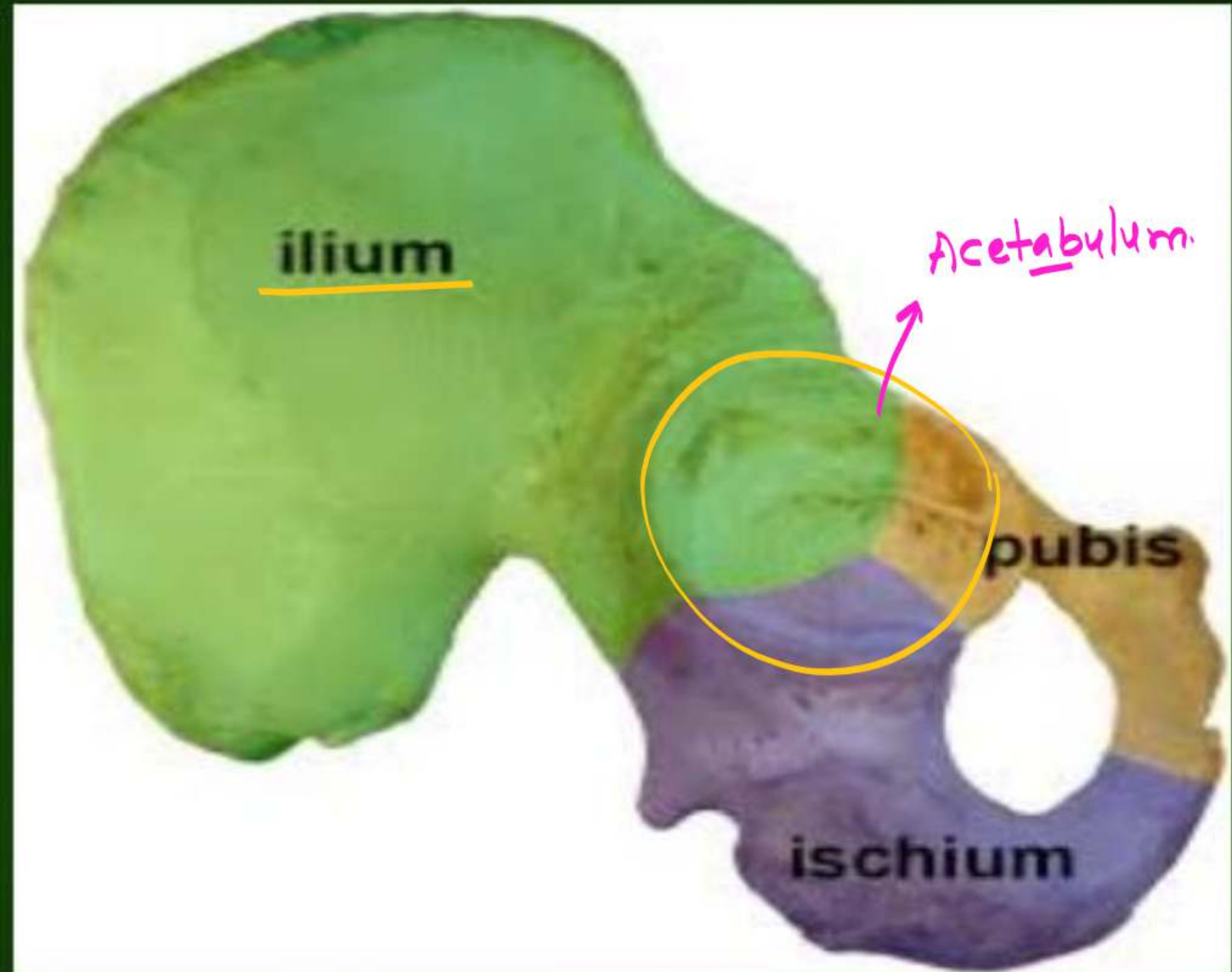
- to it Lower Limbs Attach.
- It has two halves
- Each half has 1 Bone
- Total 2 Bones
- Two- Coxal Bone

→ Each Coxal is formed by
Fusion of 3 Bones

- Ilium
- Ischium
- pubis

→ Coxal Bone has a cavity - Acetabulum
To it Head of Femur articulates

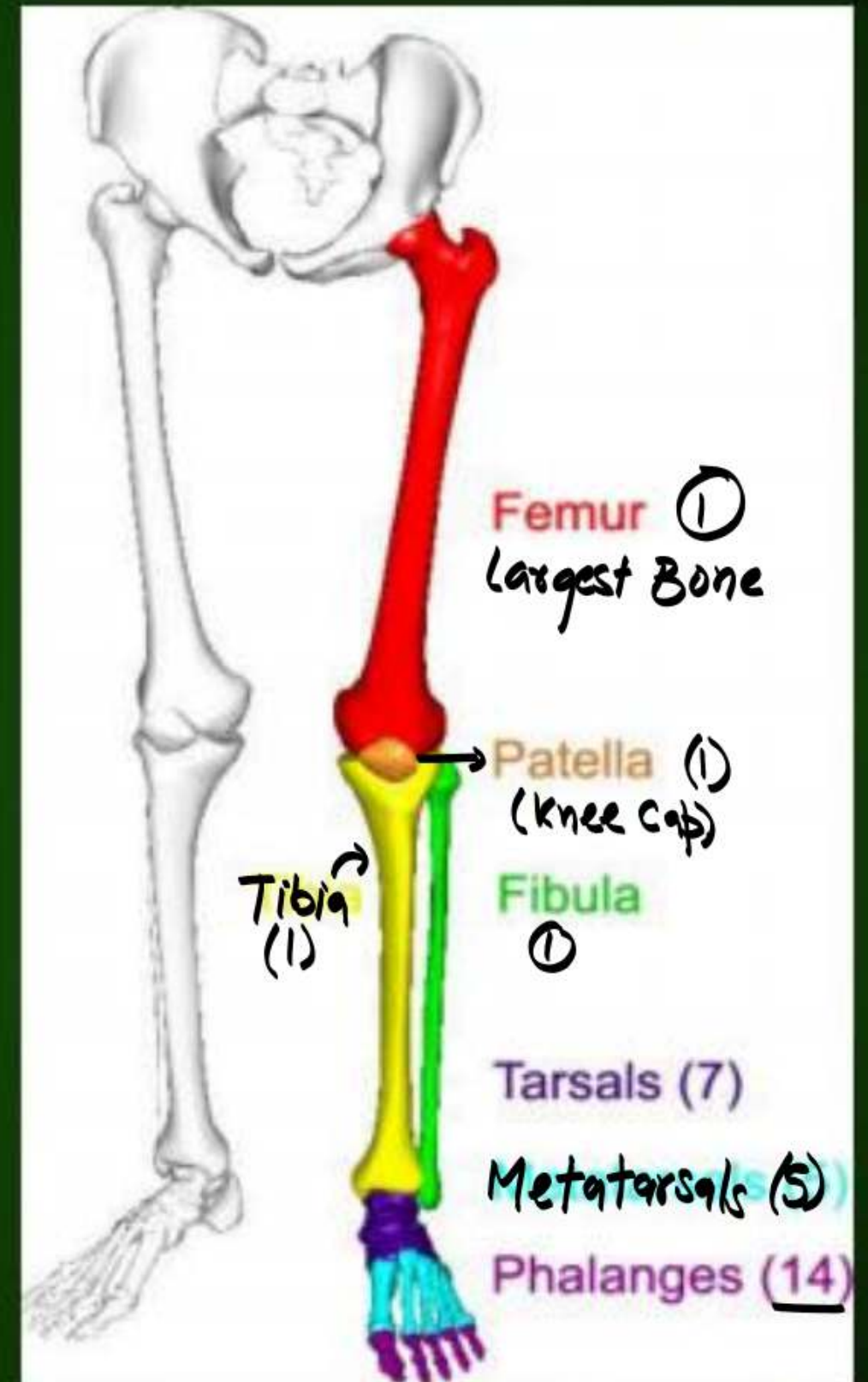
Coxal Bone



→ Lower Limb

- Each Limb Contains 30 Bones.

- Femur (1)
- Tibia (1)
- Fibula (1)
- Patella (1) → only sesamoid Bone of Body.
↓
Develops within Tendon:
- Tarsals (7)
- Metatarsal (5)
- Phalanges (14)



Joint - where two Bones Meet

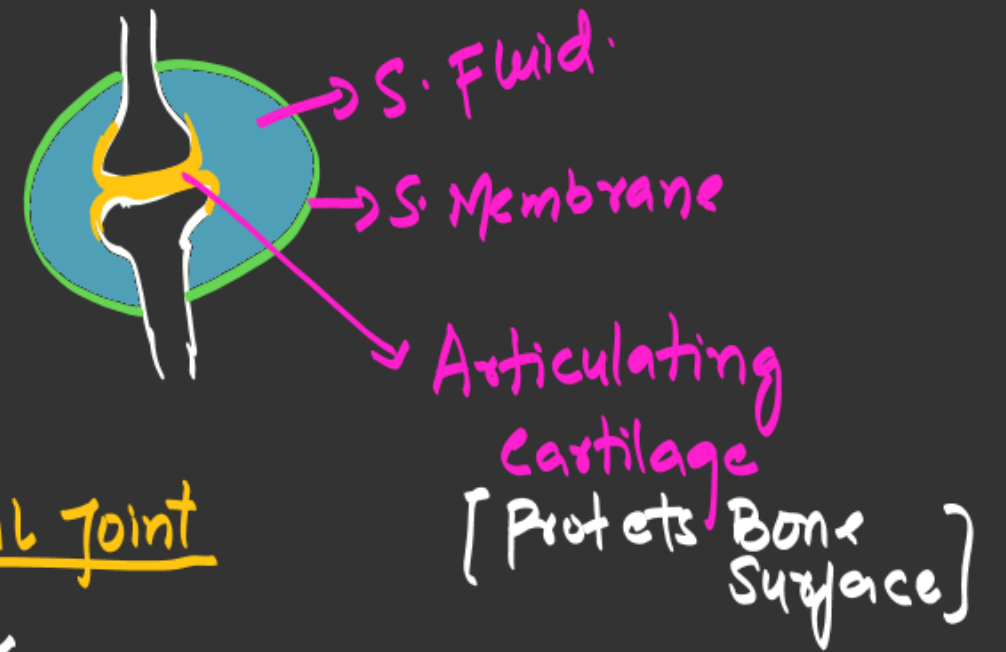
- Three types of Joints

① Fibrous Joint

↳ Immovable joints
- In Skull Bones

② Cartilagenous joint

↳ Slightly Movable
- In vertebral Column



③ Synovial Joint

↳ Movable
↳ present / covered by Synovial Membrane
↳ Inside: Synovial Fluid.

OsteoArthritis

→ Inflammation of joints due to
Damage to Articulating Cartilage with
Age or Obesity

Rheumatoid Arthritis (Autoimmune Disease)

- Immunity Against Synovial Membrane.

Examples of Synovial joints

① Ball and Socket → Shoulder joint, Hip joint

② Hinge joint → Elbow, knee joint.

③ Pivot joint → Btw Radius & Ulna.
→ Atlas (C_1) & Axis (C_2).

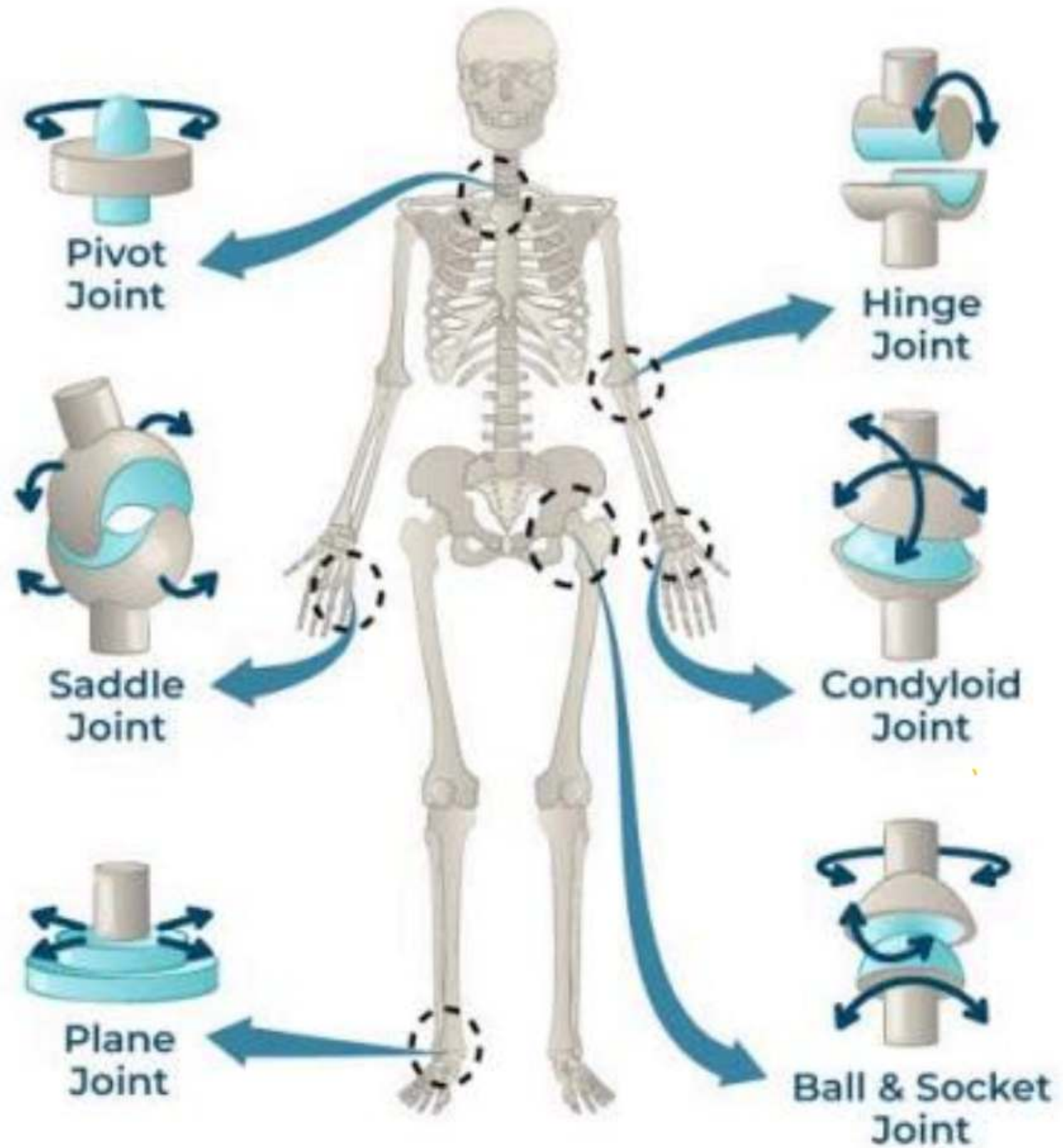
(ST) ④ Saddle joint - Between Carpel & Metacarpel of Thumb.

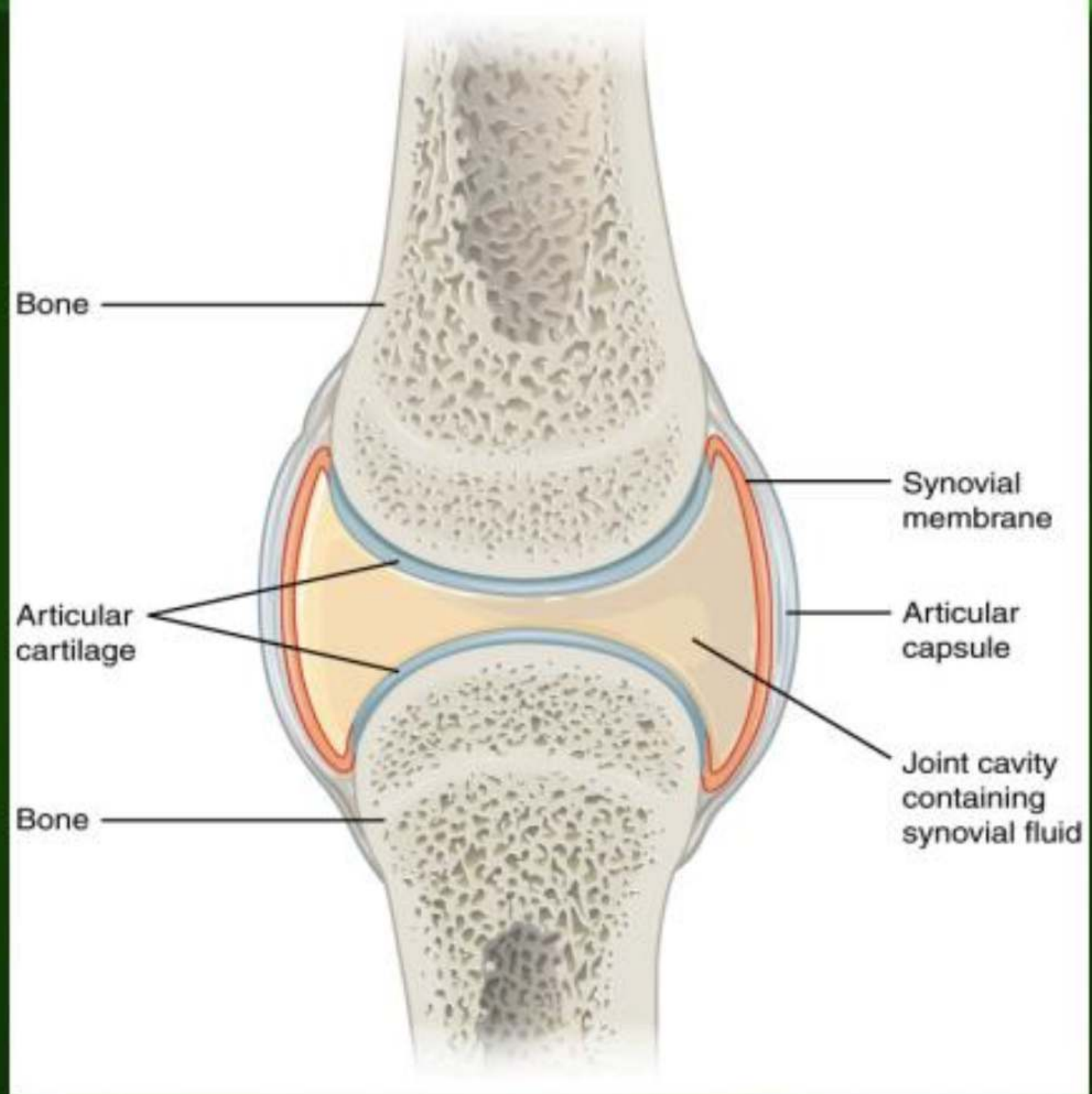
⑤ Gliding joint - Btw Carpels.

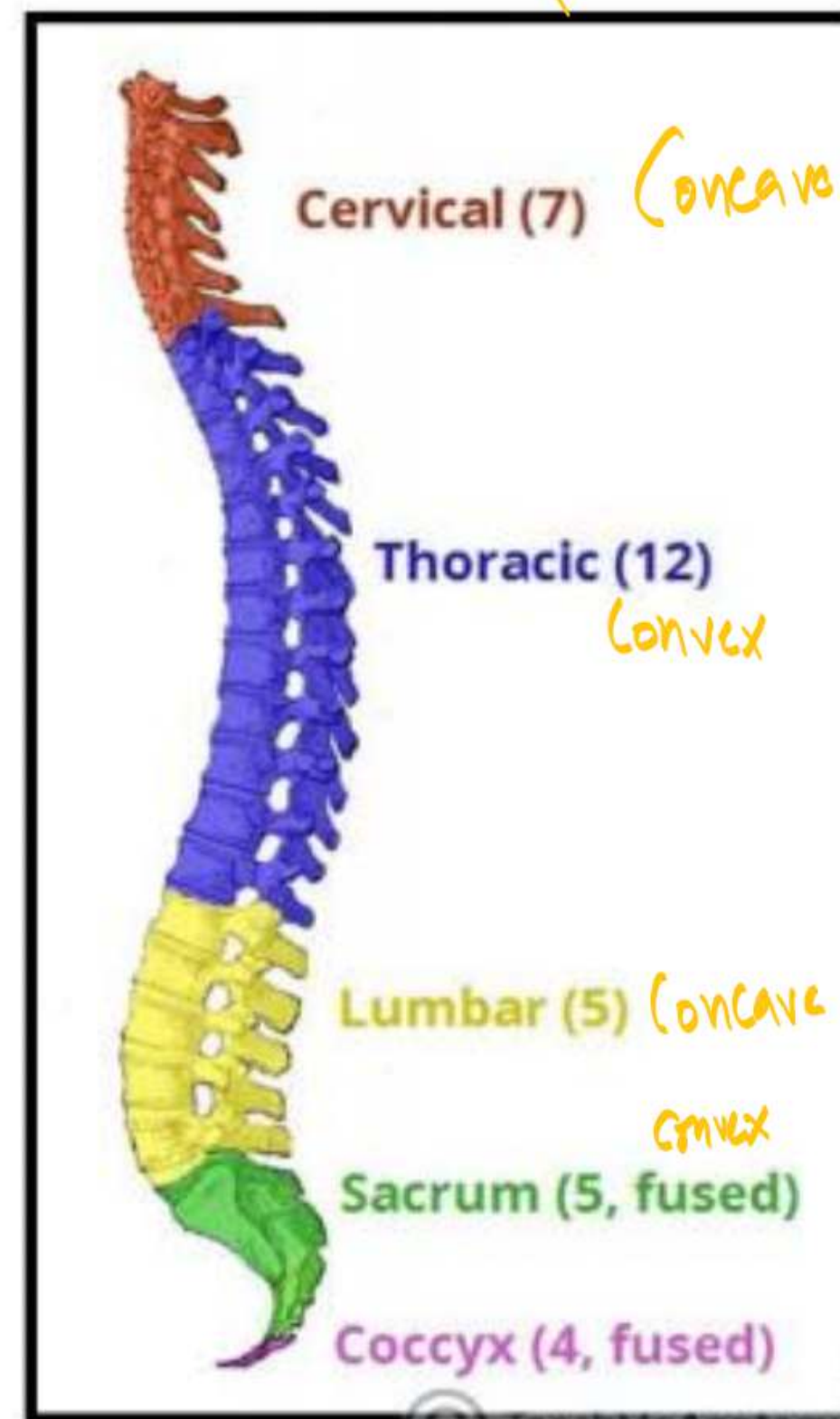
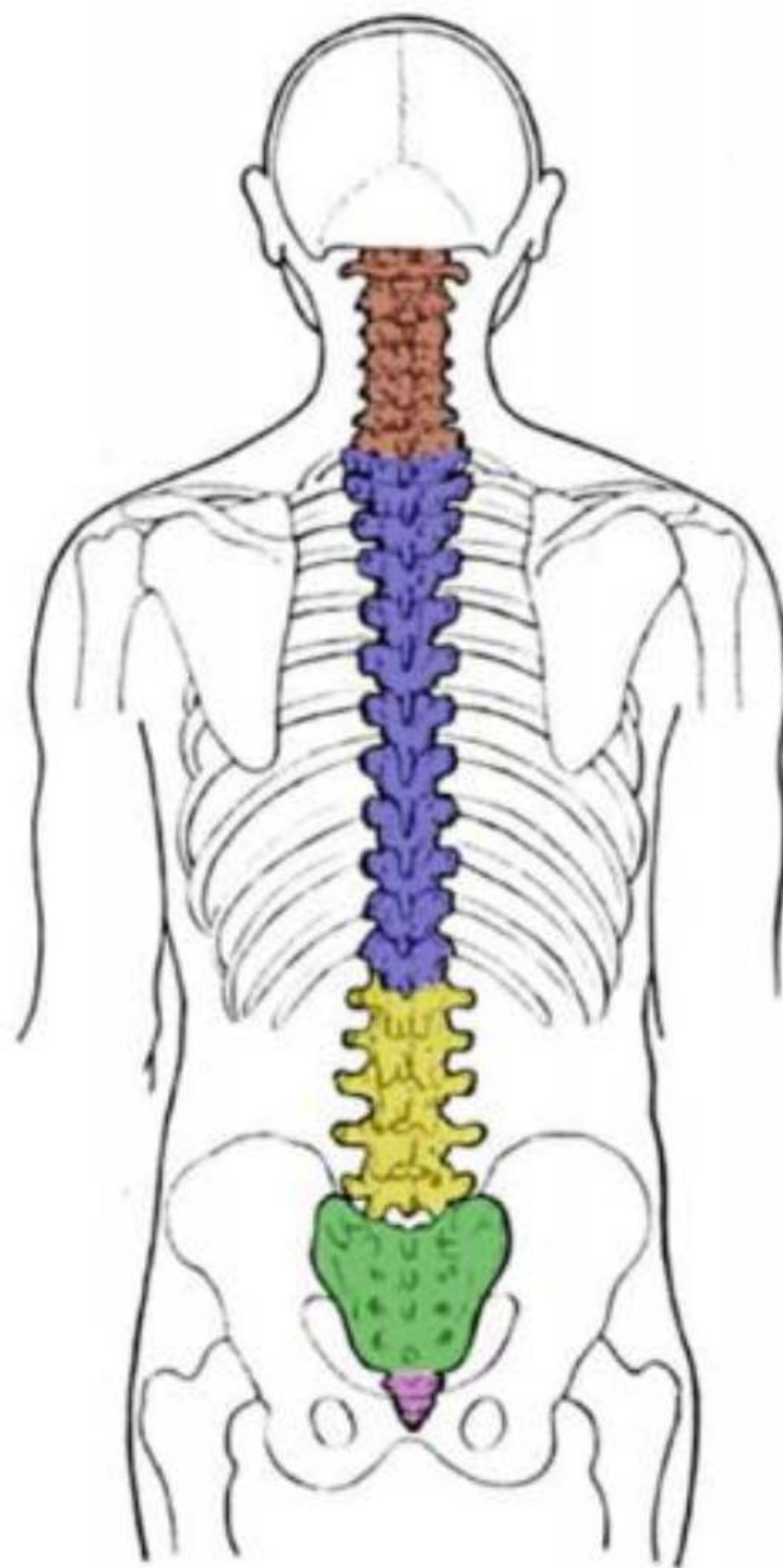
(H/W)

- Osteoporosis.
- Tetany.

Types of Joints







Thank
You

→ ③ ways
- cathly
- Mamb
- Seron